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Direct Fischer-Tropsch (Direct FT): A Novel, High Yield Synthesis Approach for Replica Fuels

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Abstract

At approximately 100 years since its invention, there is a renewed interest in Fischer-Tropsch ("FT") technology, primarily as a recognition of the need for pathways for the production of liquid hydrocarbons from waste, biomass and carbon dioxide. Established FT processes use a synthesis gas feedstock and cobalt-based catalysis in either fixed bed catalytic or slurry reactors. The process is essentially a polymerization one and the product distribution follows the Anderson- Shulz-Flory formula. This means that a broad-spectrum product distribution (in terms of molecular weight distribution) is obtained, which in turn restricts the product specificity of the process, and off-specification outputs require further processing (e.g. hydrocracking of heavy components).

At Zero Petroleum ("Zero"), we have been developing an alternative and proprietary approach we have named "Direct FT". This builds on the FT concept but has been enhanced through chemistry innovations which overcome the limitations of conventional FT. It is therefore a disruptive technology which challenges convention by delivering highly selective, tunable outputs targeted to specific fuel types such as Jet A-1, gasoline (EN228), or diesel, without post-processing or blending, and at viable small-scale production volumes. This is achieved by fine control of the molecular composition of the outputs, in particular the ratio of the PIONA (paraffinic, iso-paraffinic, olefinic, naphthenic and aromatic) components and their molecular weight distribution.

Objectives/Scope: This paper presents a breakthrough in Fischer-Tropsch ("FT") technology—"Direct FT"—developed by Zero Petroleum. Traditional FT processes, limited by cobalt-based catalysis and broad-spectrum product distributions, have restricted the commercial viability of Power-to-Liquid ("PtL" / "eFuel") fuels.

Direct FT is a disruptive technology which overcomes these barriers, challenging convention by delivering highly selective, tunable outputs targeted to specific fuel types such as Jet A-1, gasoline (EN228), or diesel, without post-processing or blending, and at viable small-scale production volumes.

Methods, Procedures, Process: Zero Petroleum's Direct FT uses a novel, proprietary catalyst system based on abundant, non-cobalt materials, engineered to yield a highly narrow carbon chain distribution. The system

allows molecular tuning of hydrocarbon output to meet exact specifications for certified fuels—particularly those requiring aromatic content—by controlling chain growth dynamics at the catalyst surface.

Unlike conventional FT that necessitates blending and hydrocracking, Direct FT operates in a simple reactor design and produces replica fuels in certifiable composition. The process has been validated in pilot-scale continuous reactors and supported by detailed analysis of product slates, distillation profiles, and compositional compliance with ASTM specifications.

Results, Observations, Conclusions: Traditional cobalt-based FT systems produce broad product distributions dominated by diesel-range paraffins and high wax content, necessitating costly upgrading (e.g. hydrocracking, separation, recycle) to generate certified fuels. These limitations significantly increase the minimum viable scale—and cost—of synthetic fuel production, hampering early-stage commercialisation of eFuels.

In contrast, Direct FT has demonstrated exceptional selectivity: fuel outputs are centered precisely in the desired molecular ranges, achieving high yields of replica Jet A-1 and EN228 gasoline with appropriate paraffinic, iso-paraffinic, naphthenic and aromatic profiles, all without post-processing. 70 litres of RON 96 aviation gasoline ("avgas"), 50 litres of Jet A-1 replica fuel and 16 litres of high-octane aromatic gasoline blend stock have been successfully synthesised.

The Jet A-1 replica candidate is undergoing ASTM certification processes. In March 2025, Zero Petroleum completed a successful flight using 100% Direct FT avgas, meeting all technical targets. These results confirm not only the operational performance of the fuels but also the viability of small, flexible Direct FT plants deployable at significantly lower capital cost. This scalability unlocks the potential for decentralised eFuel production aligned with renewable energy availability, making Direct FT a critical enabler for sustainable aviation and mobility fuels.

Introduction

Many countries have made significant progress in meeting their "Nationally-Determined Contributions" (NDCs) of greenhouse gas emissions reductions in line with UNFCCC protocols. However, certain sectors of the economy are considered to be "hard to abate", one of which is the transport industry. For example, the aviation industry contributes around 2-3% of global CO₂ emissions, and with projected growth by 2050, reducing its carbon footprint is critical. The aviation sector has limited decarbonisation options due to its reliance on liquid hydrocarbons for energy density reasons and long aircraft lifespans. Sustainable Aviation Fuels ("SAF"), including biofuels and synthetic fuels (Power-to-Liquids "PtL", electrically generated SAF "eSAF", electrically generated fuels "eFuel"), offer promising alternatives but require significant investment and policy support. Regional factors like resource availability and infrastructure play a major role in SAF deployment. A major challenge in the adoption of PtL/eSAF fuels is the current cost, estimated to be in the range of \$5-10 per litre. This is expected to reduce once processes are scaled and also benefit from improvements in renewable energy and electrolysis technology. Further improvements will arise from optimizing the selectivity and intensification of the fuel synthesis process itself, so that the yield to desired products is maximized and capital costs are reduced. This is the objective of the current study, where we focus on PtL/eFuel processes.

A typical, conventional fuel synthesis process is shown below (Figure 1); it starts with a feed of synthesis gas and produces several outputs. This is preceded by three important steps: hydrogen production through water electrolysis, carbon dioxide capture (from a process or from the atmosphere) and its conversion to carbon monoxide via the reverse water gas shift ("RWGS") reaction. Note that following the FT reactor, a large number of downstream reactions and separations are required. This is to convert the broad-spectrum hydrocarbon product of the FT reactor to products that meet established specifications.

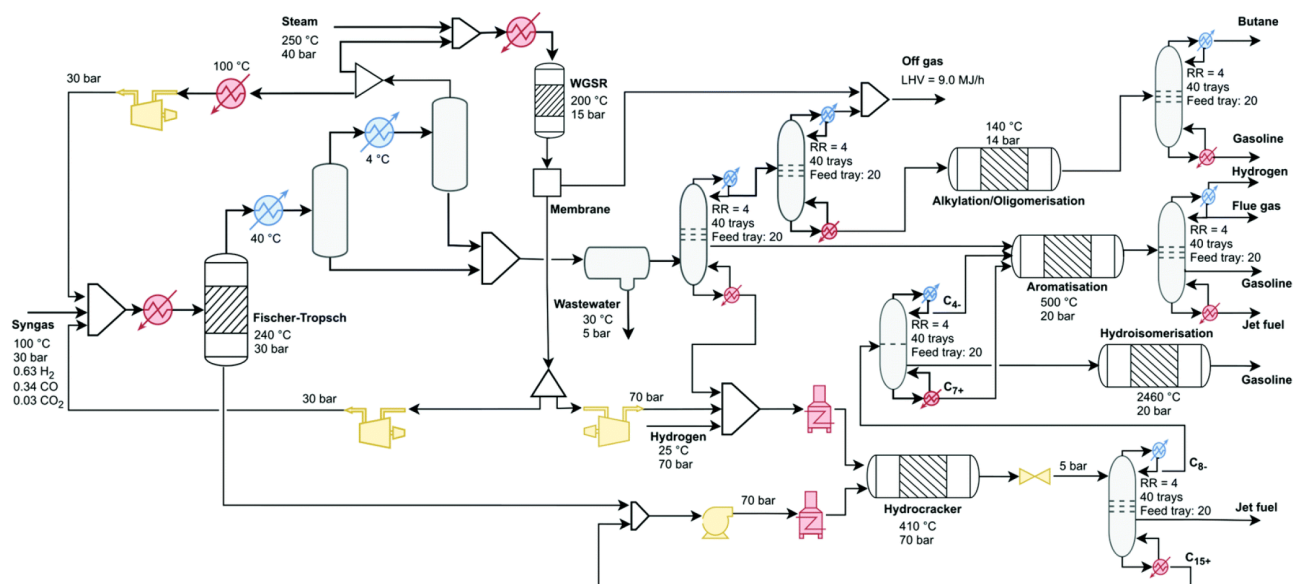


Figure 1—Flowsheet for a typical FT process (after Garay et al. 2022).

The alternative approach developed by Zero requires minimal post-processing. This is ensured by first of all operating the FT reactor under conditions to maximise the yield of the desired product (e.g. Jet A-1, EN228 gasoline, diesel, UL91 aviation gasoline or high-performance motorsport gasoline). This is followed by mild hydrogenation (to saturate any olefinic compounds) and distillation to separate products by boiling point and to recycle or separate non-product outputs. The result is a high yield of drop-in products. The process is described in more detail in the next section.

Process description

A high-level representation of the overall Zero process is illustrated below (Figure 2).

Zero's proprietary DirectFT® process

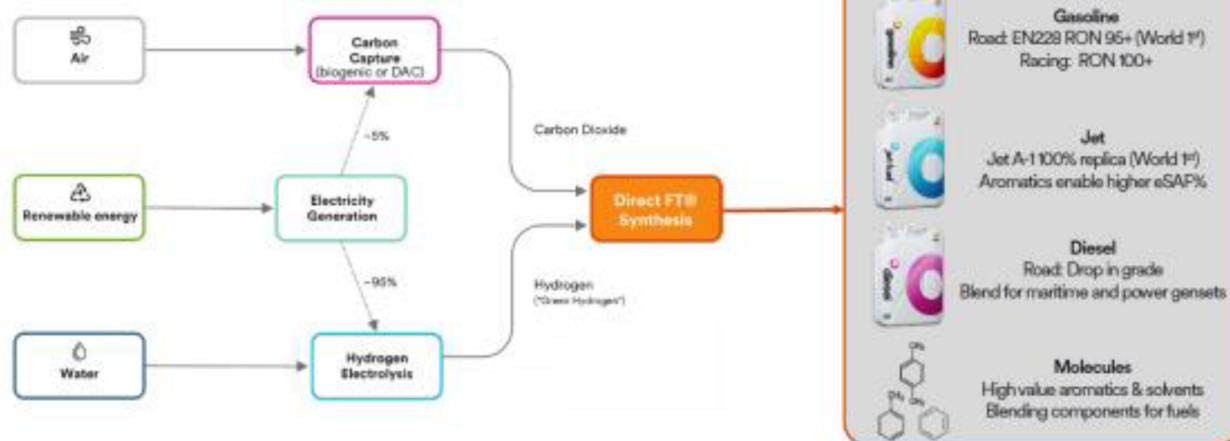


Figure 2—High-level overview of Zero synthetic fuel process

At the top level, the Zero process requires three inputs: air, water and electricity and converts these into the desired hydrocarbon fuel. The water is a source of hydrogen and the air is a source of carbon dioxide. Biogenic waste carbon dioxide can be used as well.

The core Zero process comprises three sequential reaction processes:

- i. The reverse water gas shift reaction (RWGS) which converts the carbon dioxide into carbon monoxide through a reaction with hydrogen, with a water by-product which is removed. This is now a relatively well-understood process, with a range of catalysts available and alternative technologies that can be licensed (e.g. JM Hycogen process; Topsoe e-RWGS process).
- ii. The carbon monoxide is mixed with hydrogen in the appropriate ratio for the desired product and then fed into the Direct FT reactor. In our case, the reactor is a fixed bed tubular reactor, with a heat transfer fluid on the shell side. This enables straightforward scale-up from laboratory to pilot plant to commercial plant, because once the maximum tube diameter is established, scale up occurs through increasing the size of the tube bundle in the reactor module and replication of modules.
- iii. The products of the FT reactor are hydrotreated to saturate olefins. A standard hydrogenation catalyst is used here.

Following the reaction processes, the products are separated by boiling point using distillation. This is illustrated below (Figure 3).

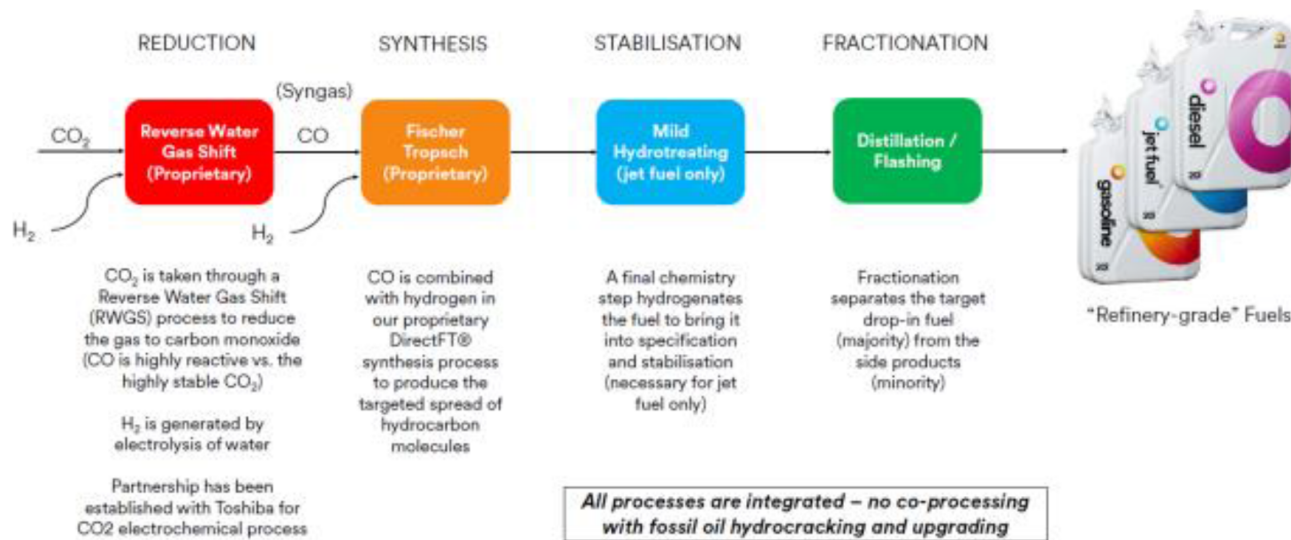


Figure 3—Illustration of Zero's sequential reaction process

The Direct FT process is the most innovative part of the overall flowsheet. Zero Petroleum's Direct FT uses a novel, proprietary catalyst system based on abundant, non-cobalt materials, engineered to yield a highly narrow carbon chain distribution. The system allows molecular tuning of hydrocarbon output to meet exact specifications for certified fuels—particularly those requiring aromatic content—by controlling chain growth dynamics and secondary reactions at the catalyst surface. Unlike conventional FT that necessitates blending and hydrocracking, Direct FT operates in a simple reactor design and produces replica fuels in certifiable composition at exceptionally high selectivity. The process has been validated in laboratory-scale continuous reactors and supported by detailed analysis of product slates, distillation profiles, and compositional compliance with ASTM specifications.

A particular feature of the process is the ability to use the reactor conditions (such as H:C ratio, temperature, pressure and residence time) to control the PIONA (paraffinic, iso-paraffinic, olefinic, naphthenic and aromatic) composition of the main product. By relating molecular composition to product properties (e.g. octane number, flash point etc.) Zero is able to design the conditions for direct production of drop-in fuels which do not need refinery post-processing or blending. The graph below (Figure 4) demonstrates how different reactor operating conditions lead to very different product compositions.

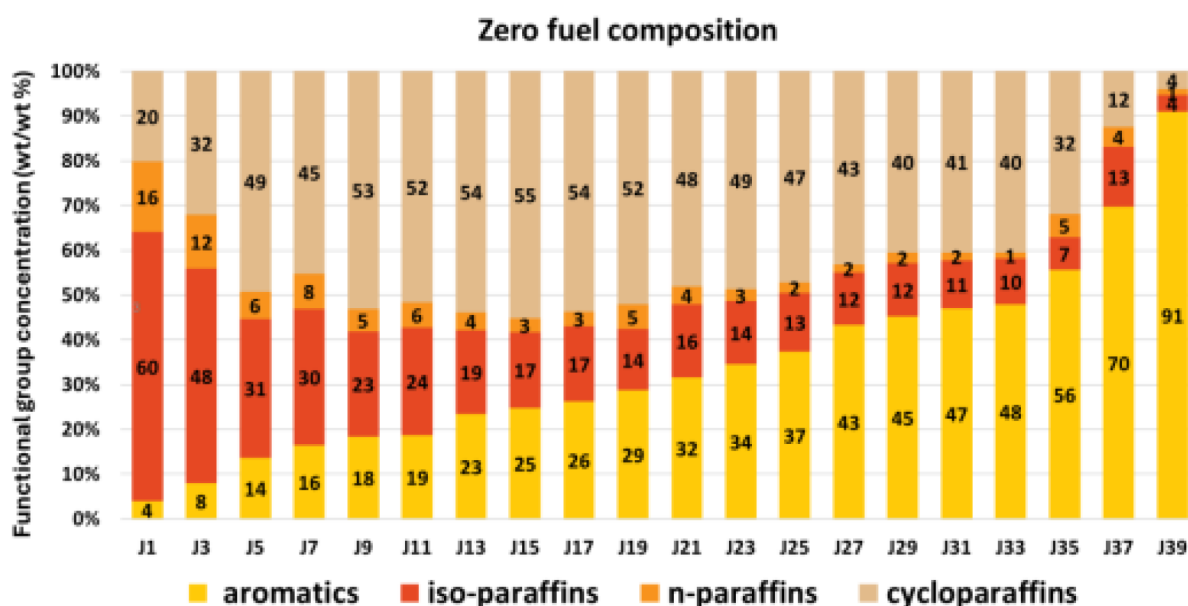


Figure 4—Demonstration of the versatility of the Direct FT process in terms of output composition.

The reaction process also has a high degree of stability, with long operating periods between catalyst regeneration possible. The results of a time-on-stream example experiment are shown below (Figure 5). The data in red shows steady fuel production over the first 15 days, followed by a drop to around 50% of the initial production rate. This drop followed a halving of the feedstock flow rate. The production rate continued steadily over the next 15 days with no sign of a drop in the rate of fuel production.

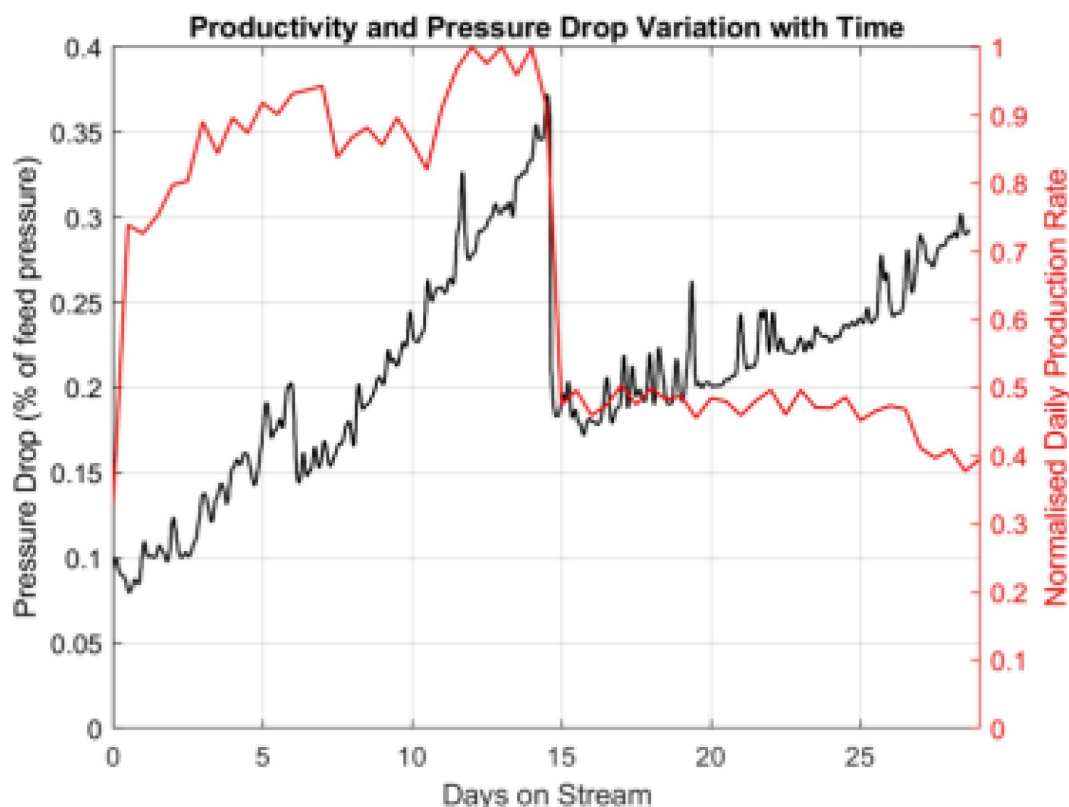


Figure 5—The normalised production rate for the FT reactor, with the pressure drop overlaid. Note that feedstock flow rates were halved on day 15

The data in black show the pressure drop across the catalyst beds, which was initially 0.1% (percent of feed pressure) and steadily increased to just below 0.4% after 15 days on stream. The change in flow rate caused the pressure drop to decrease to 0.2%, and this value increased to 0.3% over the next two weeks on-stream. The significance of the pressure drop is that it can be used to infer the formation of coke in the catalyst bed. In the current case, the pressure drop increase is very modest and indicates that the level of carbon deposition is low. When inline regeneration is required, a simple recipe is followed, and the catalyst activity is restored. It has been demonstrated that even after four regeneration cycles, catalyst activity is not affected.

An example of regeneration is shown in the graph below (Figure 6) which shows cumulative production of gasoline in a single reactor. Changes in gradient are due to adjustment in feedstock flows to ensure that fuel quality is within specification. In this case catalyst regeneration and re-activation carried out between days 30-40 before resuming production. During this run, no coking nor mechanical damage was observed. An industry standard regeneration and activation protocol was used. The time to undertake this has now been reduced significantly. Catalyst regeneration will not have a meaningful impact on plant production, as downstream process steps will be able to continue while any single reactor is being serviced. As can be observed, the production rate returns to the expected level following catalyst regeneration.

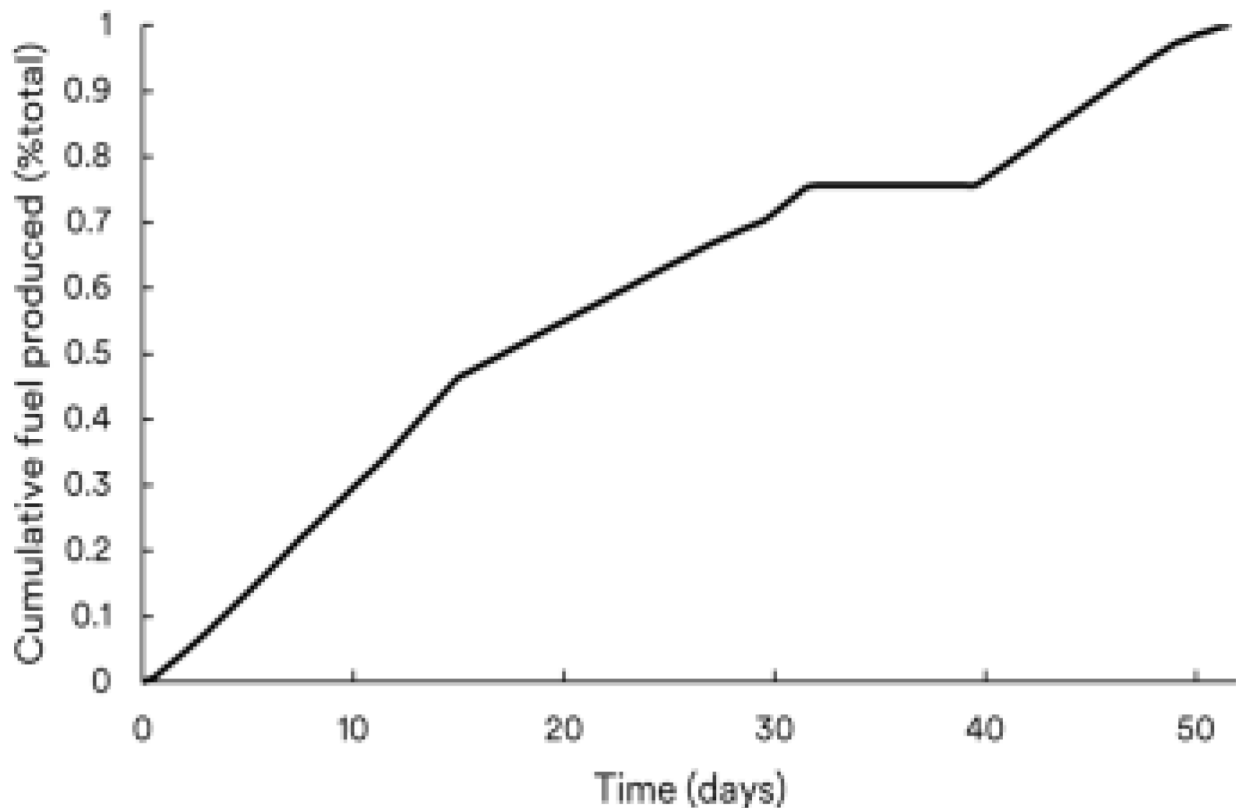


Figure 6—Long run cumulative production of gasoline - showing regeneration cycle at day 30

Key results

The ability to operate at steady conditions over a long period of time enables the production of meaningful volumes of fuel samples from Zero's pilot plants, in support of testing and certification. Direct FT has demonstrated exceptional selectivity: fuel outputs are centered precisely in the desired molecular ranges, achieving high yields of replica target 100% drop-in fuels with appropriate aromatic profiles, all without post-processing or blending.

70 litres of RON 96 aviation gasoline ("avgas"), 50 litres of Jet A-1 replica fuel and 20 litres of high-octane aromatic gasoline blend stock have been successfully synthesised. These have been manufactured separately using the same reactor without any change whatsoever in hardware. It is to be stressed, for the avoidance of doubt, that these individual 100% drop-in fuels are each produced at very high selectivity (up to 95% selectivity in liquid fraction) and not concurrently. In the anticipated commercial production, the plant can be configured to produce the single target fuel (e.g. EN228 gasoline, UL91 avgas, Jet A-1 replica, or EN590 diesel) at high selectivity according to process configuration using the same hardware.

The Jet A-1 replica candidate is undergoing ASTM certification processes at the UK SAF Clearing House in Sheffield, UK. The boiling point conformance with ASTM D4054 and World Fuel Survey Limits is illustrated below (Figure 7).

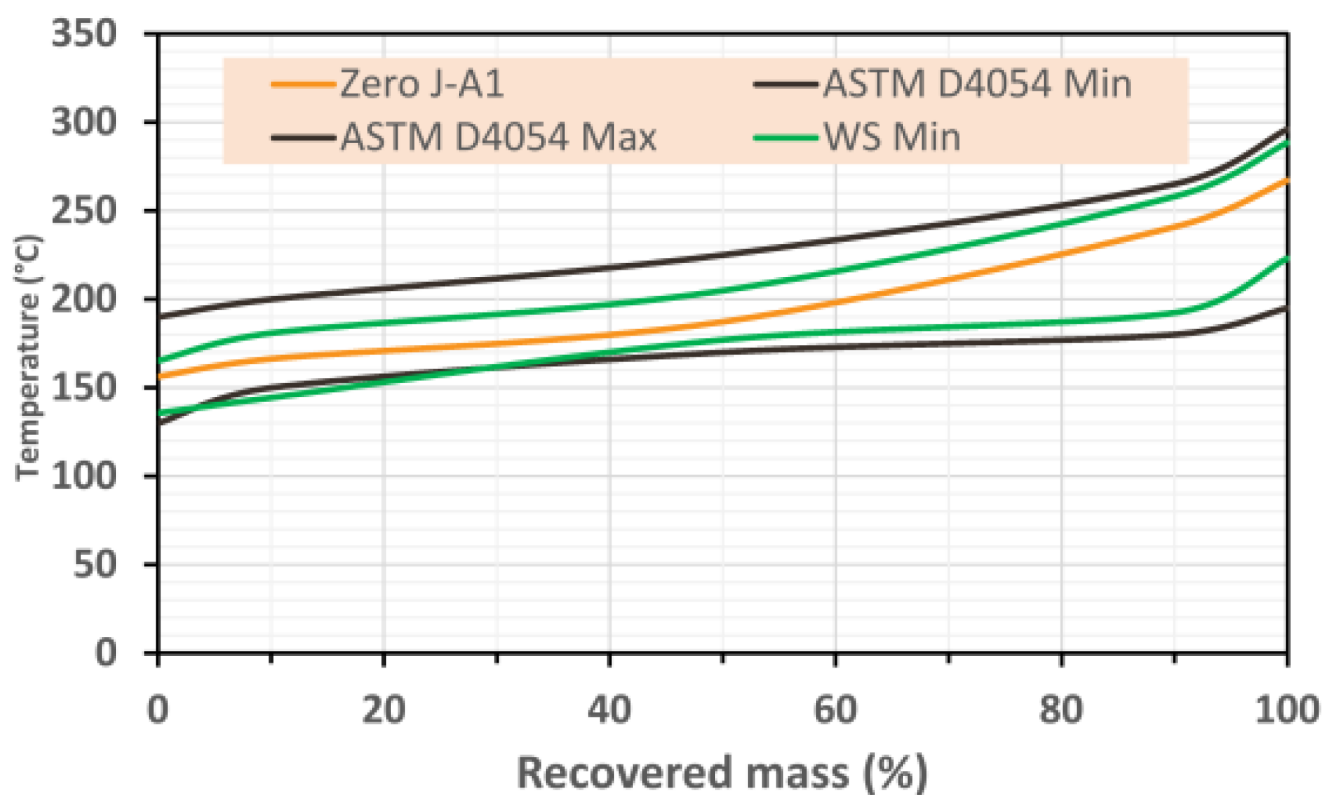


Figure 7—Boiling point distribution of Zero Jet-A compared with SATM limits and World Fuel Survey limits

In November 2022, the UK's Royal Air Force and Zero Petroleum undertook a flight using 100% Direct FT avgas and were awarded the Guinness World Records title for the first flight powered by 100% synthetic fuel (Figure 8). This was followed up in March 2025, where Zero Petroleum completed a comprehensive flight test regime using 100% Direct FT avgas, meeting all technical targets.



Figure 8—November 2022 Guinness World Records flight

Another important feature of the process is the tighter control of molecular weight distribution and low production of high molecular weight compounds ("waxes"). The latter are an issue in conventional FT processes and require additional downstream processing (hydrocracking and separation). Zero's process produces products in a relatively narrow distribution; and any components heavier than the target product can be used as a valuable by-product (e.g. marine or power generation fuel). An example of a molecular weight distribution by molecule type is shown below (Figure 9).

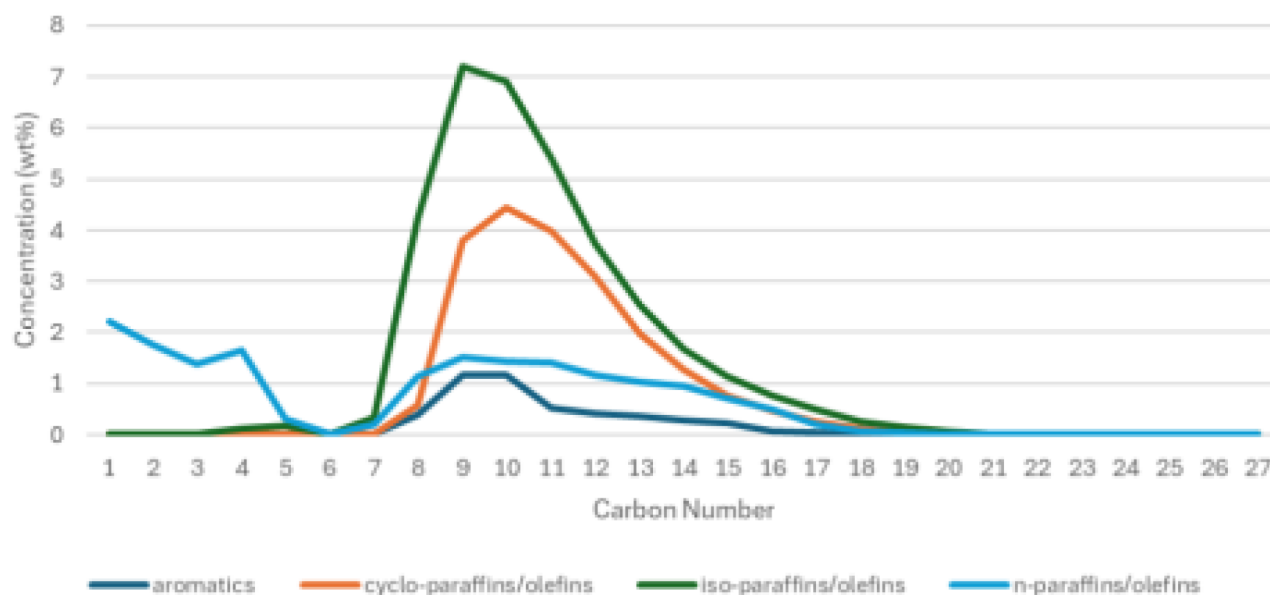


Figure 9—Carbon number distribution by component type (FT reactor output) 9

The ability to control both the component type distribution and carbon number distribution in the Direct FT process is what gives it a very high level of product selectivity and represents the core innovation of the system. Regarding yields: targets of 75-80% and 90-95% of the liquid product being within the desired specification for Jet A-1 and gasoline respectively have been demonstrated.

These outcomes confirm not only the operational performance of the fuels but also the viability of small, flexible Direct FT plants deployable at significantly lower capital cost. This scalability unlocks the potential for decentralised PtL/eFuel production aligned with renewable energy availability and likely electrolyser farm configurations, making Direct FT a critical enabler for sustainable aviation and mobility fuels. This is contrasted with the conventional approach where significant downstream processing is required which implies either significant assets directly downstream of the FT process or the need for further co-processing in a petroleum refinery.

Commercial significance

The principal performance attributes of the novel Direct FT process have been outlined as follows:

- i. Composition. Tailored PIONA synthesis capability gives the process authority to manufacture true replicas of existing fuel grades at 100% blend, or new high performance specialist fuels, and without post-processing, upgrading or blending.
- ii. Selectivity. The tunable selectivity gives exceptional productivity in the chosen target range (e.g. gasoline, jet or diesel) with the minimum of side products.
- iii. Flexibility. The same manufacturing hardware and catalysis system can manufacture the chosen target fuel, reconfigurable to a new target fuel within a matter of days.
- iv. Productivity. Preliminary analysis predicts catalysis productivity in commercial production (kinetics, in terms of hydrocarbon mass per day per catalyst mass/volume) far in excess of conventional cobalt-based Fischer-Tropsch chemistries.

These properties suggest commercial advantage in the following key aspects:

- i. Capital cost. The higher productivity requires significantly smaller reactors than conventional processes on a like-for-like basis. Capital intensive post-processing equipment such as hydrocracking reactors are not required. High selectivity to target fuels provides superior return of highest value hydrocarbon products per capital deployed. Overall Zero anticipates up to 50% lower capital costs than established FT technologies.
- ii. Operating cost and margin. Significant operational savings are made by avoiding major post-processing. In parallel to the return on capital, the high selectivity to target fuels provides superior operating margin from highest value hydrocarbon products relative to the high input costs for green hydrogen and carbon dioxide feedstocks.
- iii. Minimum viable scale. The ability to produce 100% drop-in fuel grades without post-processing, refining or blending allows manufacture of final products without associated partnerships such as refineries for blend stock or upgrading by mass-balance or otherwise. This allows even micro-scale Direct FT plants to operate commercially. The global scale-up of PtL/eFuel production requires access to significant renewable power resources, ideally without breaching additionality principles. Renewable power is itself constrained in the rate of scale-up. The minimum viable scale of Direct FT will enable much faster and earlier traction with lower minimum viable power demand opening a much wider range of project locations and lower quantum of capital demand for the plants themselves.
- iv. Logistics and storage. 100% replica fuels can be shipped direct to client without further logistics to refinery or blending facilities. Due to their inherent compatibility, these fuels can be stored alongside

- existing fuel grades and/or mixed entirely with existing fuel stores, pipelines and distribution vehicles for sale on an offset basis to remote clients. New distribution assets are not required as they are for non-fungible fuels such as those produced by other candidate synthetic or biofuel chemistries.
- v. Access to markets. The same plant and the associated deployed capital can make multiple fuels, reconfigurable to the commercial market favoured at the time. The compositional authority opens the market to not only conventional fuel grades but also to advanced and bespoke fuels for premium clients.
 - vi. Territorial opportunity. There are multiple territories without access to refineries or blend stocks for completion of unfinished fuels. Direct FT provides finished fuels and is therefore viable for direct-to-client production in remote locations such as island nations and communities, and indeed very much larger territories such as Australia which have limited refinery capacity.

Conclusions

Concawe has predicted 80Mtoe as the eFuels production volumes in 2050 (https://www.concawe.eu/wp-content/uploads/Rpt_21-7.pdf), a figure higher than that expected from lignocellulosic residues and wastes. Typical pathways for such fuels are via alcohols (e.g. the methanol to jet and ethanol to jet processes) or via FT synthesis. The latter has gone through various stages of development and the paradigm developed for gas-to-liquids technology has been demonstrated to benefit from large scale plants. This is particularly the case because of the significant amount of post-processing required following the FT reaction stage (see Figure 1). This has not been an important issue because these processes (e.g. the Pearl and Oryx plants) are located where natural gas supplies are plentiful and economies of scale are easy to achieve.

For future PtL/eFuel plants, we expect the scale to be limited by a number of factors including availability of renewable power (with plant sizes of order 100-300 MWe input likely) and carbon dioxide. This means that process economics will only partially be improved through scaling, after which process optimisation will be important. The conventional FT process will need to follow one of two pathways: (i) crude fuel synthesis followed by co-processing in a separate petroleum refinery (with credits derived from mass balance calculations) or (ii) downstream processing in a directly-adjacent facility.

Due to its ability to produce 100% drop-in products at high selectivity, product-configurable using the same hardware, and with minimal downstream processing, Direct FT represents a step-change in synthetic fuel production. By eliminating the need for refining infrastructure and enabling narrow-range, high-purity fuel synthesis directly from synthesis gas, it delivers a flexible, efficient pathway for renewable hydrocarbon fuels. The technology enables right-sized, modular PtL/eFuel deployment and redefines economic thresholds for synthetic fuel production, accelerating the transition to net-zero mobility.

These positive and disruptive attributes of Direct FT in composition, selectivity, flexibility and productivity have been considered from a commercial perspective, identifying significant prospective advantages in capital expenditure, operating expenditure and margin, minimum viable scale, logistics, access to markets and territorial opportunity.